

Quick Check

Lesson 6: Final Explanation: Controlled Growth, Metamorphosis and Survival in Kenyan Insects

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.2: Growth and Development in Animals | Lesson 6

A graph shows the body length of a grasshopper nymph over time. The line rises sharply at some points and stays flat between them. What best explains the flat sections?

- A** The nymph stops eating and starts to shrink
- B** The nymph has become an adult and has stopped moulting
- C** The hard exoskeleton cannot stretch, so the nymph cannot grow bigger until it moults
- D** Hormones are completely absent from the body during these times

During one step (moult) phase, a grasshopper nymph grows from 14 mm on Day 10 to 20 mm on Day 13. Using growth rate = change in length $\div$ time taken, what is its growth rate in this phase?

- A 6 mm per day
- B 2 mm per day
- C 6.7 mm per day
- D 0.5 mm per day

In a caterpillar, which statement about ecdysone and juvenile hormone is correct?

- A** Juvenile hormone triggers moulting, and ecdysone keeps the insect in the larval form
- B** Both hormones stop the caterpillar from moulting
- C** Ecdysone makes the exoskeleton stretch so no moulting is needed
- D** Ecdysone triggers moulting, and a high level of juvenile hormone makes the moult produce another larval stage

Which pair correctly matches each insect with its life-cycle stages?

A Butterfly: egg, larva, pupa, adult. Grasshopper: egg, larva, pupa, adult

B Butterfly: egg, nymph, adult. Grasshopper: egg, nymph, adult

C Butterfly: egg, nymph, adult. Grasshopper: egg, larva, pupa, adult

D Butterfly: egg, larva, pupa, adult. Grasshopper: egg, nymph, adult

A caterpillar eats sukuma wiki leaves, while the adult butterfly drinks nectar from flowers. How does this difference help the species survive?

- A** The larva and the adult do not compete for the same food
- B** The caterpillar can use nectar when leaves run out
- C** It lets the butterfly avoid growing and moulting
- D** The butterfly no longer needs any food after the pupa stage

A student finds an empty, split insect skin on a maize leaf beside a larger grasshopper. Which statement is the best scientific explanation?

- A** The grasshopper peeled its skin like sunburned skin because it was damaged by the sun
- B** The grasshopper melted inside its skin and rebuilt a new body without any control
- C** The grasshopper shed its old rigid exoskeleton during moulting, triggered by ecdysone, and a new larger exoskeleton replaced it
- D** The skin split because the grasshopper stretched it smoothly as it grew

During the Driving Question Board sorting, which question can the class place under 'Answered (with evidence)' using evidence from this unit?

A Why does the growth curve of an insect have steps instead of a smooth line?

B How might human pesticides disrupt hormonal pathways in agricultural pests?

C Do other animals, such as crabs in Mombasa, moult like insects?

D Which new chemicals could block ecdysone in crop pests?

In their Day 1 versus Final comparison, a learner writes about why a calf and a grasshopper nymph grow differently. Which statement is correct?

- A** Neither animal needs hormones to control growth
- B** A calf grows in a sigmoid (S-shaped) pattern without an exoskeleton limit, while an insect grows in steps because its exoskeleton must be replaced
- C** A calf grows in steps and an insect grows smoothly because insects are smaller
- D** Both grow in steps because both have a hard exoskeleton

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).